Grading on a Curve? An Analysis of NYC Restaurant Health Inspection Scores

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1 Introduction

New York City grades every restaurant in its five boroughs through the Department of Health and Mental Hygiene (DOHMH) inspection system. Each inspection assigns a numerical score based on health code violations found on the day of inspection, where a lower score indicates better performance. A score of 13 or below earns an "A," 14 to 27 earns a "B," and 28 or above earns a "C." These letter grades are publicly posted at restaurant entrances, making the grading system unusually visible and consequential for businesses and consumers alike.

In 2014, data journalist Ben Wellington published an analysis noting an anomalous spike in inspection scores at exactly 13, the highest score that qualifies for an A grade [1]. Wellington's explanation was that inspectors may exercise informal leniency near grade boundaries, perhaps because a score of 13 is judged equivalently to a score of 10 by the letter-grading rubric, while a score of 14 carries a substantial reputational cost. The NYC Health Department denied this, stating that "inspectors are not instructed to offer leniency." Wellington's data, however, only covered 2010 to 2014, and no one has formally revisited the question with newer data.

This study does exactly that, using inspection data updated through 2026. We extend the original analysis in four directions. First, we formally test whether the grade-boundary spike persists and measure how large it has become. Second, we quantify whether a restaurant's borough and cuisine type explain meaningful variation in inspection scores, using a regression framework and a two-way blocked ANOVA. Third, we examine whether violation types are distributed consistently across the four boroughs, or whether geography predicts which kinds of food safety hazards are most prevalent. Finally, we assess how well pre-inspection information can actually predict whether a restaurant will earn an A grade, providing an honest upper bound on what contextual factors alone can tell us.

The public health stakes are real. Firestone and Hedberg (2018) showed that the NYC grading program reduced Salmonella infections by 5.3% per year relative to the rest of New York State following its 2010 implementation. Understanding what drives variation in inspection outcomes is therefore not only a statistical question but a public health one.

2 Data

2.1 Sources

The primary dataset is the DOHMH New York City Restaurant Inspection Results file, downloaded from NYC OpenData on April 3, 2026. It contains 113,571 records spanning inspections from 2007 through early 2026. The data is structured with one row per violation per inspection, so a single inspection event appears multiple times if multiple violations were cited. Key variables include borough, cuisine type, inspection score, inspection date, violation code, and inspection type (initial cycle, re-inspection, reopening, etc.).

Violation codes alone are not self-explanatory. We used a secondary NYC Health Department reference file mapping each code to a descriptive category such as "Pest Control," "Cold Holding," or "Handwash / Toilet." We also merged neighborhood-level American Community Survey estimates at the 2020 Neighborhood Tabulation Area (NTA) level, including median worker earnings and total population, to capture the socioeconomic context of each restaurant's location.

2.2 Preparation

Several preprocessing decisions were made to ensure the analysis is methodologically sound. Because the raw data has one row per violation, we deduplicated to one row per inspection event whenever computing scores or temporal trends, taking the first score value within each restaurant-date pair. This prevents inspections with many violations from being weighted more heavily than inspections with fewer.

Staten Island was excluded throughout due to sparse cell counts in the balanced ANOVA design. Inspections with missing scores were removed. The cleaned dataset contains 101,840 records covering 19,065 unique restaurants across four boroughs.

Cuisine types were grouped into ten *hypercategories* to reduce dimensionality and improve interpretability. For example, Chinese, Japanese, Korean, and related cuisines were collapsed into "East Asian." The Dietary Style category was also excluded from the ANOVA due to sparse borough-level cell counts. Table 1 summarizes the dataset.

Table 1: Dataset summary after cleaning.

Statistic	Value
Inspection records	101,840
Unique restaurants	19,065
Boroughs	Bronx, Brooklyn, Manhattan, Queens
Full data coverage	2022–2026
A-grade rate	38.7%
Mean score (SD)	25.5 (19.3)
Median score	21

A note on temporal coverage: while the dataset nominally spans 2007 through 2026, the vast majority of inspection records fall in 2022 through 2025. Prior to 2022, fewer than 90 inspection events per year appear after deduplication, making those years unsuitable for inference. All temporal analyses are therefore restricted to 2022 through 2025.

3 Methods

3.1 Grade-Boundary Spike

To test whether the spike at score 13 is statistically significant, we compared observed inspection counts across scores 12 to 21 against an expected flat distribution. The expected count per score was estimated as the mean count at scores 14 to 21, a range safely removed from both grade cutoffs. A chi-square goodness-of-fit test was applied to observed versus expected counts. This approach follows Wellington's original methodology but formalizes it with an inferential test.

We replicated the analysis separately for each borough to examine whether the boundary leniency varies geographically.

3.2 Temporal Trend

We restricted to initial-cycle inspections between 2022 and 2025 and aggregated to one observation per inspection event. Mean score by year was computed for each borough, and we estimated the overall trend using ordinary least squares regression of mean score on year. We also tracked A-grade rates by year and borough to see whether the probability of earning a top grade has shifted over this period.

3.3 OLS Regression of Inspection Score

To assess the simultaneous contributions of borough, cuisine type, and time on inspection score, we fit an ordinary least squares regression:

$$\text{Score}_i = \beta_0 + \boldsymbol{\beta}_{\text{boro}} + \boldsymbol{\beta}_{\text{cuisine}} + \beta_{\text{year}} \cdot t_i + \varepsilon_i$$

where borough and cuisine are entered as categorical indicators (Bronx and African cuisine as baselines) and year is centered at 2022. We used inspection-level data, deduplicated to one row per inspection event. Confidence intervals are reported at the 95% level. The regression quantifies how much each factor contributes to score variation, and crucially, how much variation remains unexplained.

3.4 Two-Way Blocked ANOVA

To complement the regression with a formal testing framework, we fit a two-way ANOVA with interaction:

$$\text{Score}_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where α_i represents the borough effect and β_j represents the hypercategory effect. To maintain a balanced design and prevent large sample sizes from producing trivially small p-values for negligible effects, we sampled $n = 30$ observations per hypercategory-borough cell.

3.5 Violation Type by Borough

To answer whether violation types are uniformly distributed across boroughs, we built a contingency table of violation category by borough and applied Pearson's chi-square test of independence. We also computed the percentage of inspections in each borough that received each violation type and displayed these rates as a heatmap.

3.6 Violation-Level Logistic Regression

Rather than trying to predict grades from pre-inspection characteristics alone, we asked an inferential question: given that a violation was cited during an inspection, does the type of violation predict whether the restaurant will earn an A? This is a legitimate inferential question because different violations carry different point values and may reflect different levels of underlying hygiene risk.

We fit a logistic regression with violation category, borough, cuisine hypercategory, and neighborhood income as predictors of the binary A-grade outcome. The coefficient plot for violation categories provides a controlled ranking of which violation types are most strongly associated with failing an A.

3.7 Predictive Logistic Regression

As a final analysis, we assessed how well a restaurant can be predicted to earn or fail an A grade using only information available before inspection: borough, cuisine hypercategory, neighborhood median earnings, neighborhood population, and year. Predictors were standardized on the training set. We used an 80/20 stratified train-test split and evaluate performance using AUC and accuracy on the held-out test set.

4 Results

4.1 The Grade-Boundary Spike Is Massive and Consistent

Figure 1 shows the full score distribution. The spike at 13 is visually striking. Two grade cutoffs are apparent: one at 13 (A/B boundary) and a smaller one at 28 (B/C boundary). Both are consistent with inspectors rounding near boundaries.

The chi-square test is conclusive. At scores 12 and 13, we observe 17,863 inspections against an expected 3,033 under a flat distribution. This is an excess of 14,830 inspections, or +489% above what would be expected under fair grading ($\chi^2 = 30{,}792$, $p \approx 0$). Wellington's 2014 finding has not only persisted but appears to have intensified with more data.

Figure 2 breaks the spike down by borough. The excess at scores 12 to 13 is large in every part of the city: Bronx (+359%), Brooklyn (+497%), Manhattan (+504%), and Queens (+519%). Notably, Manhattan and Queens show the largest proportional excesses, suggesting that boundary leniency is not more restrained in wealthier or more heavily scrutinized areas.

4.2 Inspection Scores Have Worsened Since 2022

Among the 2022–2025 initial-cycle inspections where data is sufficiently dense, mean score increased from 16.3 in 2022 to 19.0 in 2025. An OLS regression of annual mean score on year estimates the rate of increase at 0.89 points per year ($R^2 = 0.96$, $p = 0.022$). Figure 3 shows the trend with 95% confidence intervals.

Despite the rising mean, the median score hovers around 12 to 13 throughout this period, which corresponds directly to the grade boundary spike identified in Section 4.1. The mean rises because the upper tail of the distribution is spreading, but the median is anchored near 13 by the concentra-

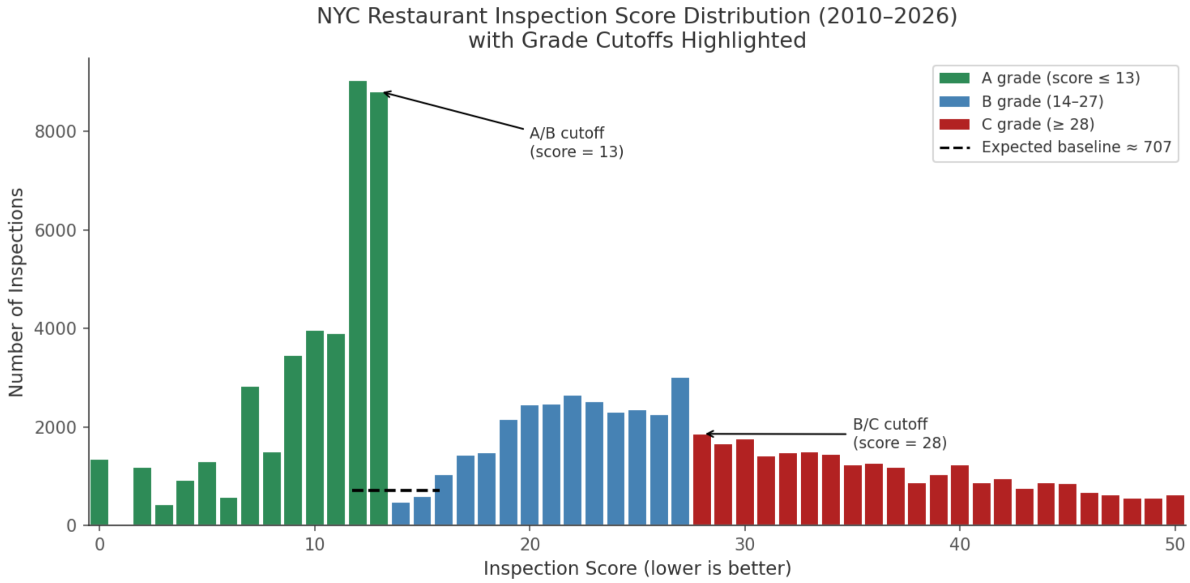


Figure 1: Distribution of NYC restaurant inspection scores across all inspection types (2010–2026). Green bars are A-grade inspections (score ≤ 13), blue are B-grade, and red are C-grade. The dashed black line marks the expected flat baseline near the A/B cutoff, estimated from scores 14–21.

tion of scores just below the grade cutoff.

Figure 4 shows A-grade rates by borough from 2022 to 2025. Manhattan consistently has the highest A-grade rate (roughly 60 to 63%), while Queens has the lowest (dropping to about 52% in 2025). All boroughs show a modest declining trend over this period.

4.3 Borough and Cuisine Effects Are Real but Modest

The OLS regression of inspection score on borough, cuisine, and year yields an R^2 of just 0.036. Even with year controlled for, only 3.6% of the variance in inspection scores is explained by borough and cuisine type. This means that the vast majority of variation reflects factors specific to individual restaurants rather than their location or cuisine.

Within this limited explanatory power, some patterns are worth noting. After controlling for cuisine type and year, Queens scores are significantly higher (worse) than the Bronx baseline by 1.64 points ($p < 0.001$), and Brooklyn by 0.86 points ($p = 0.038$). Manhattan, counterintuitively, is not significantly different from the Bronx once cuisine composition and year are accounted for ($\beta = 0.12$, $p = 0.77$). The apparent advantage of Manhattan restaurants in raw comparisons likely reflects their cuisine mix rather than better sanitation practices. Figure 5 shows the full coefficient plot.

Cuisine hypercategory is a stronger predictor than borough. African cuisine restaurants serve as the baseline and have the highest expected scores; all other groups score significantly or near-significantly lower. Beverages & Sweets establishments score about 9 points better than African cuisine restaurants on average ($p < 0.001$), and American restaurants score about 8.4 points better ($p < 0.001$). These are substantive differences in absolute terms, though the total R^2 remains low because within-category variation is still large.

The two-way blocked ANOVA ($n = 30$ per cell) confirms these patterns formally. Cuisine hypercategory is significant ($F(9, 1160) = 4.12$, $p < 0.001$) and the interaction between cuisine and borough is also significant ($F(27, 1160) = 1.54$, $p = 0.038$), indicating that the cuisine effect on score differs across boroughs. Borough alone is only marginally significant ($F(3, 1160) = 2.25$, $p = 0.081$). The full ANOVA table is in Appendix B.

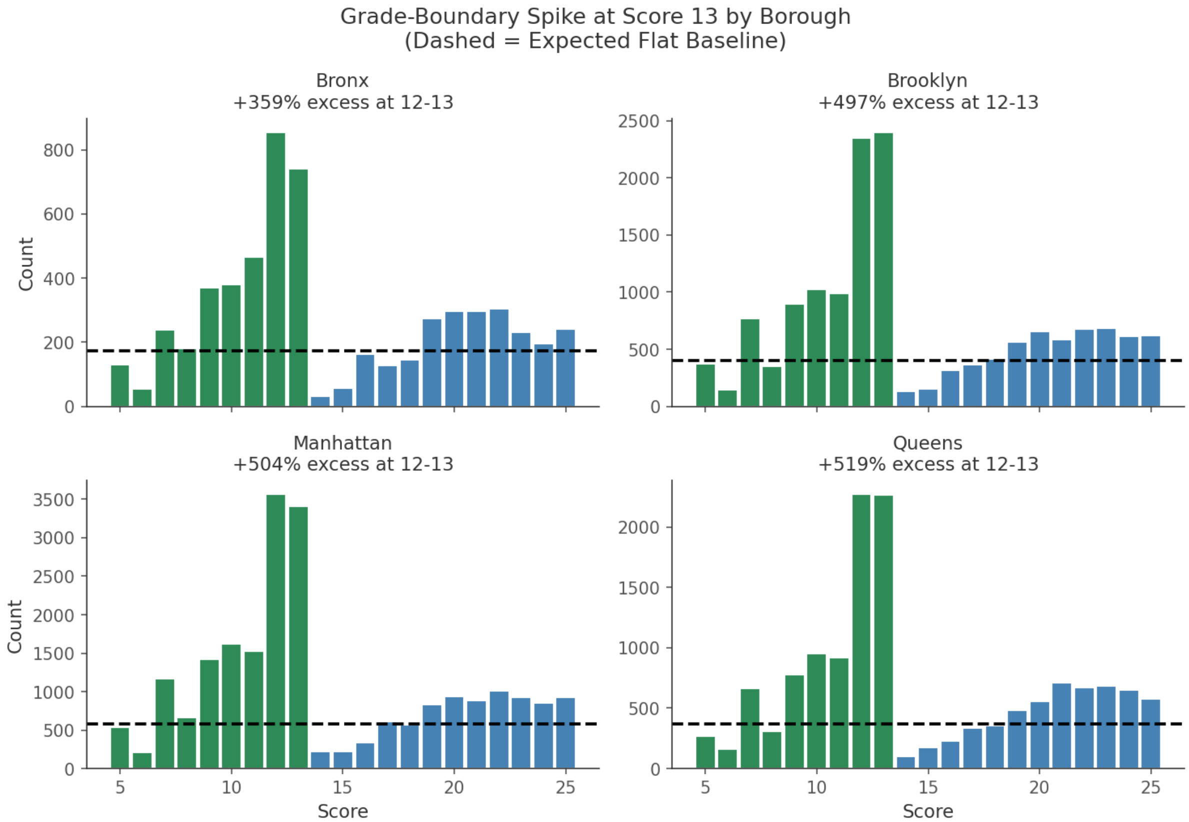


Figure 2: Grade-boundary spike at score 13, replicated separately for each of the four boroughs. The dashed line shows the expected flat baseline. All boroughs show a substantial excess at scores 12–13, ranging from +359% (Bronx) to +519% (Queens).

4.4 Violation Types Vary Substantially Across Boroughs

Research Question 3 asked whether certain violation types are more common in some boroughs than others. The chi-square test of independence strongly rejects the null hypothesis of no association between violation type and borough ($\chi^2 = 678.2$, $df = 66$, $p \approx 2 \times 10^{-102}$). This is an extremely large test statistic, reflecting genuinely different violation profiles across the city.

Figure 6 shows the proportion of inspections in each borough that received each violation category. Several patterns stand out. Pest Control violations are most concentrated in the Bronx relative to other boroughs. Cooling and Refrigeration violations are elevated in Manhattan. Queens and the Bronx show notably higher rates of Hot Holding violations. Food Workers violations (concerning food handler hygiene) are elevated in Brooklyn. These patterns likely reflect a combination of building infrastructure differences, restaurant type composition, and neighborhood characteristics.

4.5 Which Violations Are Most Associated with Failing an A?

Figure 7 shows the logistic regression coefficients for each violation category, controlling for borough, cuisine, and neighborhood income. All violation types with negative coefficients are associated with a lower probability of earning an A grade, meaning they are associated with higher inspection scores.

Pest Control violations have the largest negative coefficient, meaning that receiving a Pest Control citation during an inspection is the most strongly associated with not earning an A grade. Permit violations (PERMIT/FPC) and Handwash/Toilet violations follow. Temperature-related violations, including Cold Holding and Hot Holding, are also

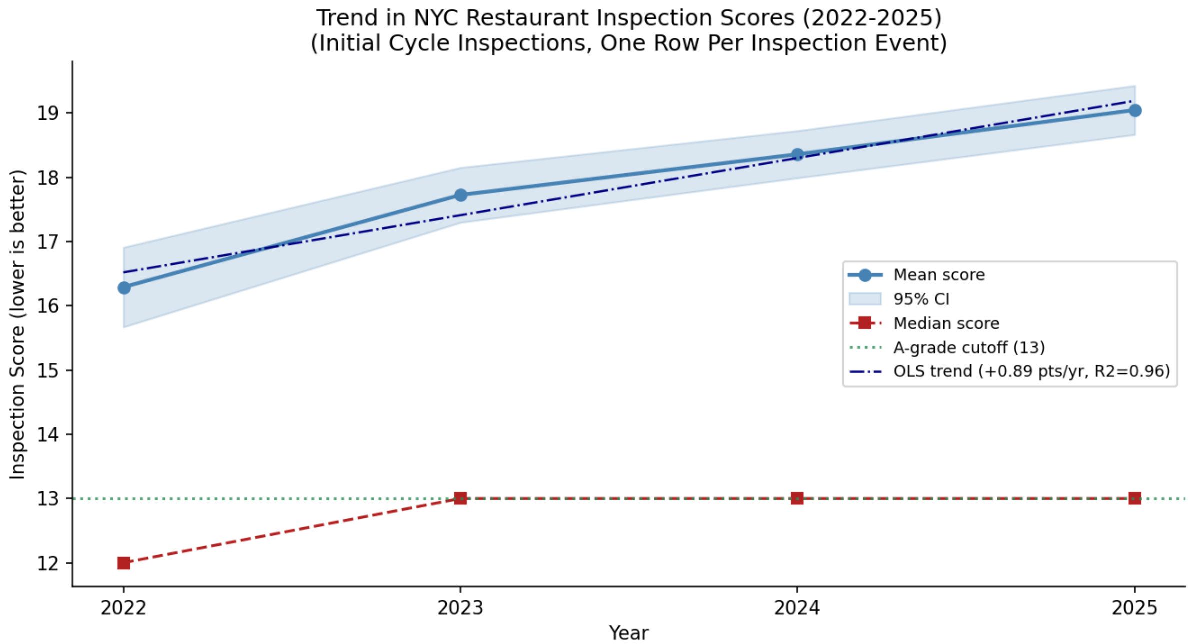


Figure 3: Mean and median inspection score by year (2022–2025), initial cycle inspections only, one row per inspection event. Shaded region is a 95% CI on the mean. The green dotted line marks the A-grade cutoff at 13. The dash-dot line is the OLS fit.

consistently negative. On the other end, violations related to Maintenance, Construction & Placement and Plumbing are actually positively associated with earning an A grade. This pattern makes sense: structural and plumbing violations are often lower-point items that inspectors cite even at well-run establishments, whereas temperature and pest violations carry higher point values and signal more serious food safety failures.

The violation-level logistic model achieves an AUC of 0.673, meaning it correctly ranks a randomly chosen failing inspection above a randomly chosen passing inspection about 67% of the time. This is a reasonable result for a logistic model predicting a complex outcome; it demonstrates that violation type is meaningfully informative about whether a restaurant will pass, even after controlling for cuisine and neighborhood.

4.6 Pre-Inspection Context Cannot Predict Grade

The pre-inspection logistic regression, using only borough, cuisine, neighborhood income, neighborhood population, and year, achieves an AUC of 0.60 and an accuracy of 59.4% on the held-out test set. Both figures are barely above a naive baseline classifier. Figure 8 shows the ROC curves for both the violation-level model and the pre-inspection model side by side.

Table 2 summarizes performance for both models.

Table 2: Logistic regression model performance on held-out test sets.

Model	AUC	Accuracy
Pre-inspection features only	0.60	59.4%
With violation type	0.673	67.2%

The gap between the two models (AUC 0.60 vs. 0.67) represents the additional information contained in what violation types were actually cited. That gap exists but is modest, which reinforces the main message: inspection outcomes are driven primarily by what happens inside a restaurant on a given day, not by where the restaurant is located or what kind of food it serves.

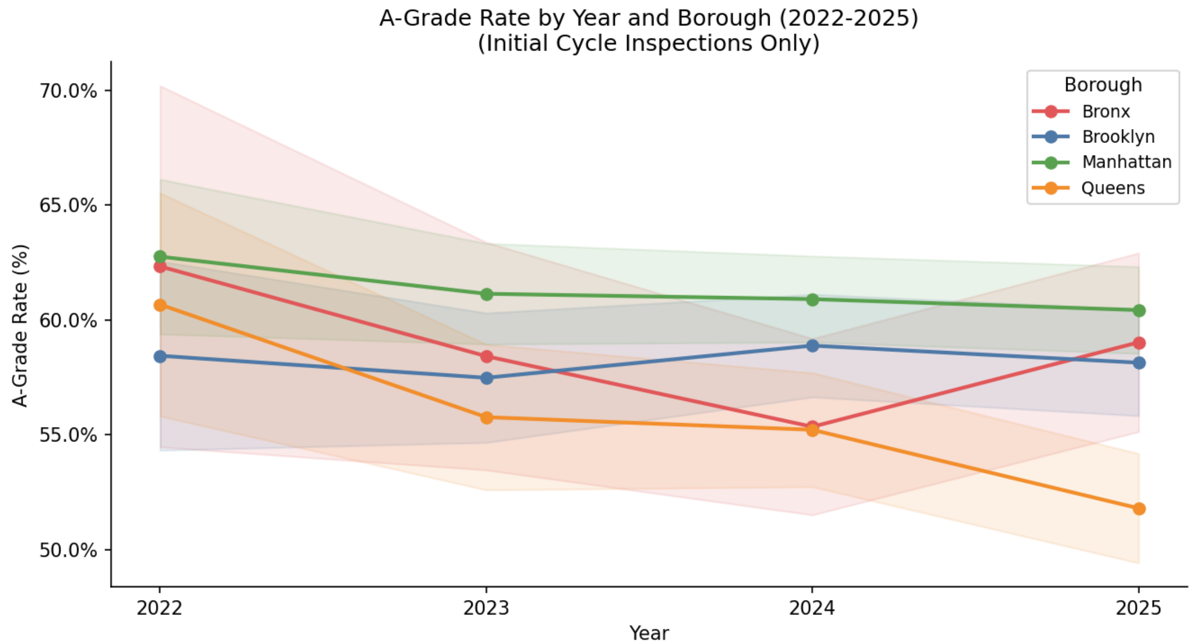


Figure 4: A-grade rate (score ≤ 13) by year and borough, 2022–2025, for initial cycle inspections. Shaded bands are 95% confidence intervals on the proportions. Manhattan leads throughout; Queens declines most sharply.

5 Model Validation

Both logistic regressions were evaluated on stratified 80/20 hold-out splits, with standardization fit only on training data, ensuring no information from the test set influenced model training or preprocessing. The reported AUC values are honest out-of-sample estimates.

For the ANOVA, balanced sampling ($n = 30$ per cell) was chosen deliberately. With over 100,000 inspection records in the dataset, fitting the ANOVA on the full data would produce near-zero p-values for substantively tiny effects due to the extremely high power of large-sample tests. The balanced design forces interpretations to rely on effect magnitudes, which are modest, rather than p-values alone.

The OLS regression reports $R^2 = 0.036$. This is not a modeling failure but a substantive finding: borough, cuisine, and year together explain less than 4% of the total variance in inspection scores. The remaining 96% reflects within-group variation, driven by factors specific to individual restaurants and individual inspection days that our data does not capture.

The chi-square tests for the bump analysis and the violation-by-borough analysis are extremely well-powered at this sample size. The observed effect sizes in both cases, a 489% excess at the grade boundary and a chi-square statistic of 678 for violation-by-borough, are substantive enough to be visible in raw counts and not statistical artifacts.

6 Conclusion

Four findings emerge from this analysis.

The grade-boundary spike documented by Wellington in 2014 remains large and statistically unambiguous. With data through 2026, we observe nearly 15,000 excess inspections at scores 12 and 13 relative to what a uniform distribution would predict, an excess of 489%. This pattern holds in every borough. Despite the Health Department's assurances to the contrary, the data continues to suggest that inspectors exercise leniency near the A/B boundary, and the spike at the B/C cutoff (score 28) provides additional corroboration.

Inspection scores are worsening. Among initial-

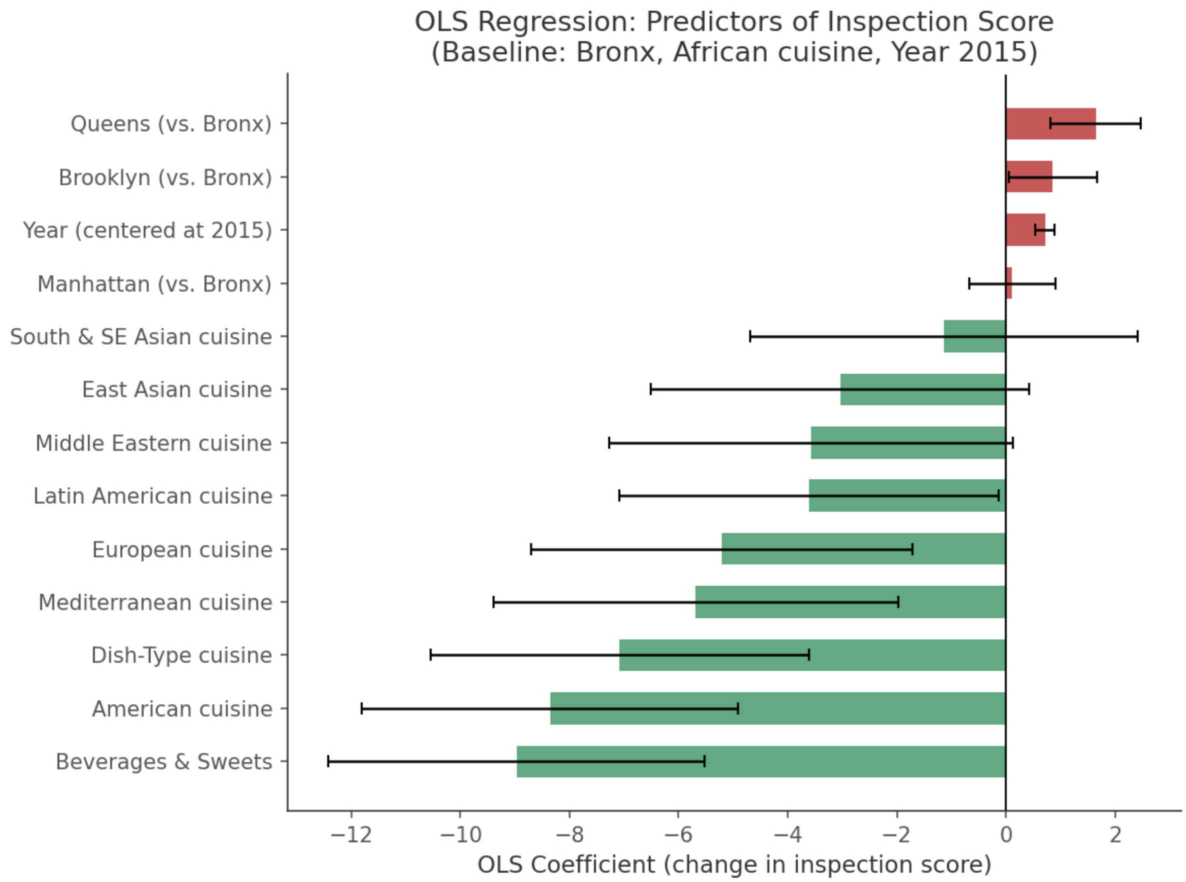


Figure 5: OLS regression coefficients for inspection score. Red bars indicate higher (worse) scores relative to the baseline (Bronx, African cuisine, year 2022); green bars indicate lower (better) scores. Error bars are 95% confidence intervals. Year has a consistent positive effect even after controlling for other factors.

cycle inspections from 2022 to 2025, the mean score increased by roughly 0.9 points per year. The A-grade rate has declined modestly across all four boroughs. Whether this reflects actual deterioration in restaurant sanitation, changes in inspection stringency, or shifts in the restaurant population is an important open question that the available data cannot answer on its own.

Violation types are not geographically uniform. Different boroughs show distinct violation profiles, with Pest Control violations elevated in the Bronx, temperature-related violations more concentrated in Manhattan and Queens, and Food Workers violations elevated in Brooklyn. These differences are highly statistically significant and suggest that food safety interventions targeted by borough and violation type might be more effective than uniform city-wide approaches.

Pre-inspection context has little predictive power. Knowing a restaurant's borough, cuisine, and neighborhood income allows us to correctly classify its inspection outcome only about 60% of the time. This is barely better than guessing. Even adding violation type only brings AUC to 0.67. Restaurant-level practices and management quality, the things we cannot observe in this dataset, appear to be the primary drivers of inspection outcomes.

Future work could examine whether the grade-boundary spike varies by inspector team or time of year, model temporal trends at the restaurant level using fixed effects to separate individual restaurant trajectories from city-wide shifts, or link inspection records to consumer review data to assess whether lower grades actually translate to lower revenue and behavioral change.

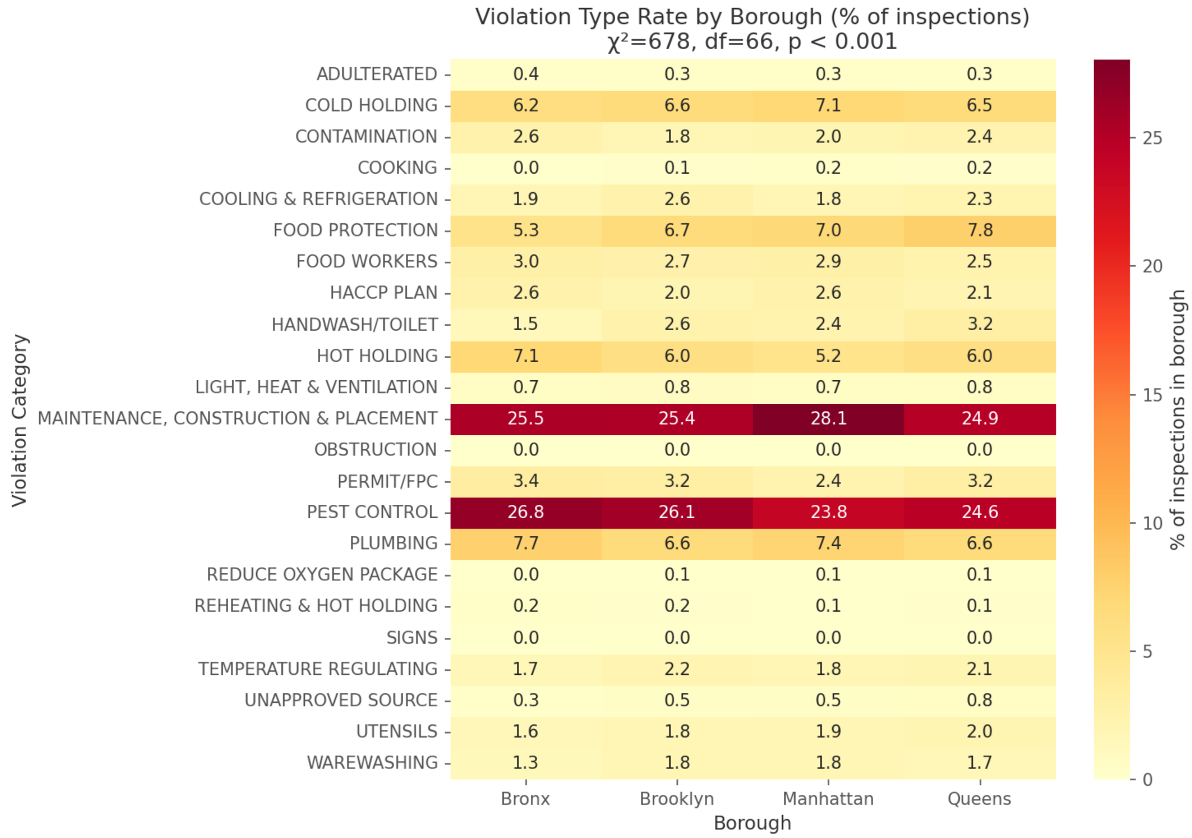


Figure 6: Violation category rate by borough, expressed as the percentage of inspections in that borough that received each violation type. Darker shading indicates higher rates. The chi-square test strongly rejects independence ($\chi^2 = 678.2$, $df = 66$, $p \approx 2 \times 10^{-102}$).

A Literature Review

Our project was motivated by two bodies of work. The first is Wellington's (2014) original analysis identifying the spike at score 13. The second is the public health literature evaluating whether these grades actually affect health outcomes.

Firestone and Hedberg (2018) examined Salmonella infection rates in NYC versus the rest of New York State from 1994 to 2015, using a quasi-experimental interrupted time-series design. They found that following the letter-grading program's launch in July 2010, NYC's Salmonella rates declined by 5.3% per year relative to the rest of the state (IRR $= 0.95$, 95% CI: 0.92–0.98, $p < 0.01$). Before 2010, NYC rates were significantly higher than the state average; by 2011–2015, the two were statistically indistinguishable. The authors are careful about causal attribution, noting that the grading program coincided with other policy changes including increased inspection frequency and higher fines. They also point to ecological study limitations: case counts are based on residence, not on where individuals ate. Nevertheless, their finding provides strong motivation for our project. If better grades are associated with fewer foodborne illnesses at the population level, then understanding what drives variation in grades becomes a question with direct public health consequences.

Their paper also highlights a concern directly relevant to our predictive modeling: the grade itself is an aggregate summary of many underlying violation-level decisions. By modeling violation-level detail, we aim to decompose what the grade reflects, which is what Firestone and Hedberg treat as a black box.

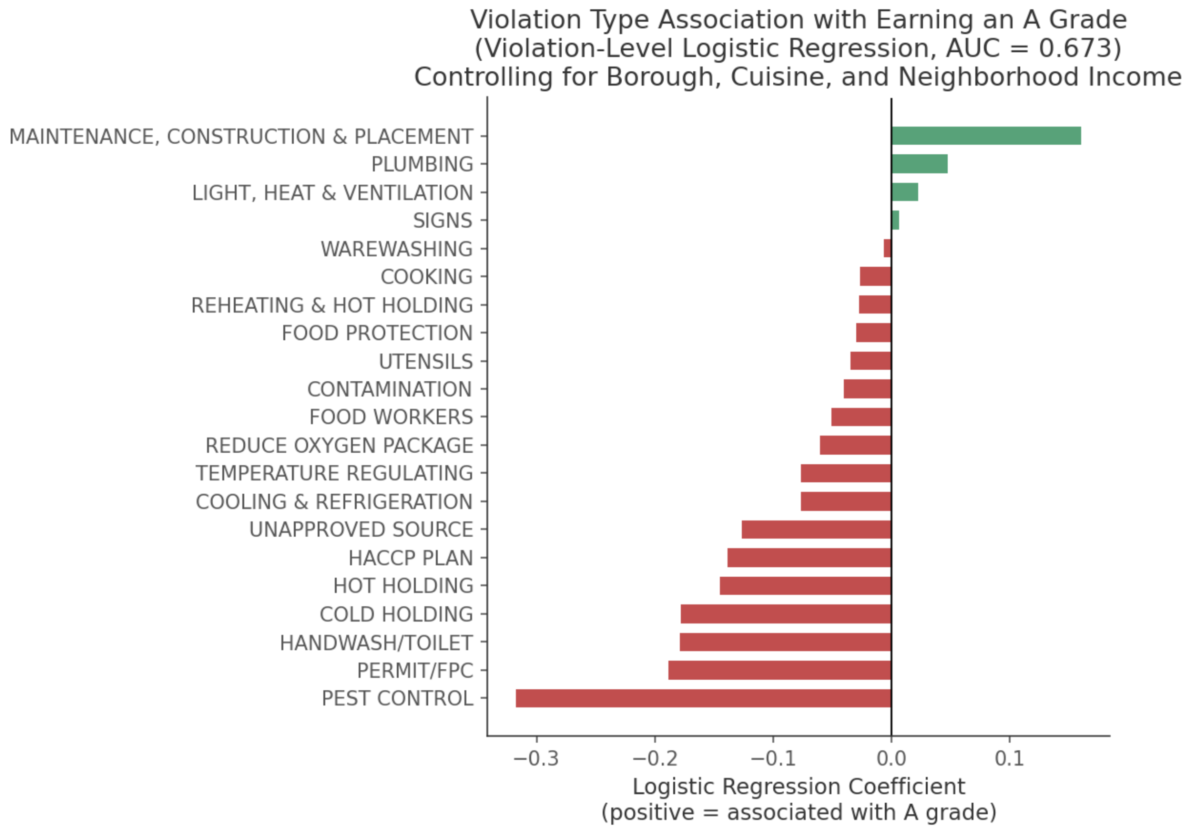


Figure 7: Logistic regression coefficients for violation category, predicting probability of earning an A grade. Red bars indicate violation types associated with failing an A (negative effect on A-grade probability); green bars indicate violation types that restaurants earning A grades are more likely to have received. Model AUC = 0.673.

B ANOVA Table

Table 3: Two-way ANOVA results. Balanced sample with $n = 30$ per hypercategory-borough cell. Baseline: Bronx borough, African cuisine. Staten Island and Dietary Style hypercategory excluded.

Source	Df	F	p	
Hypercategory	9	4.12	< 0.001	***
Borough	3	2.25	0.081	.
Hypercategory:Borough	27	1.54	0.038	*
Residuals	1160			

Signif.: *** $p < 0.001$, * $p < 0.05$, . $p < 0.1$

C Additional Figures

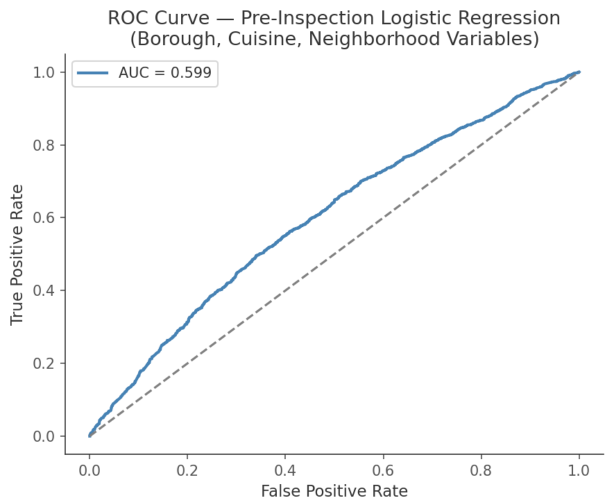
D Code

Python analysis code is in `new_analysis.py` and `extended_analysis.py`. The R analysis and ANOVA are in `group_analysis.Rmd`. All files are submitted alongside this report.

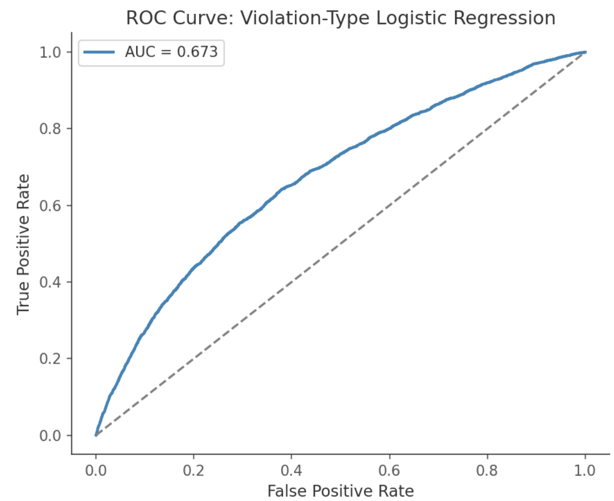
References

[1] Wellington, B. (2014). *Think NYC Restaurant Grading is Flawed? Here's the Proof.* I Quant NY. https://iquantny.tumblr.com/post/76928412519

[2] Firestone, M. J. & Hedberg, C. W. (2018). Restaurant inspection letter grades and Salmonella infections, New



(a) Pre-inspection model (AUC = 0.60).



(b) Violation-level model (AUC = 0.673).

Figure 8: ROC curves for the two logistic regression models. Even knowing what type of violations were cited (right) adds substantial information beyond pre-inspection context alone (left). Neither model achieves high AUC, consistent with the finding that most score variation is idiosyncratic to individual restaurants.

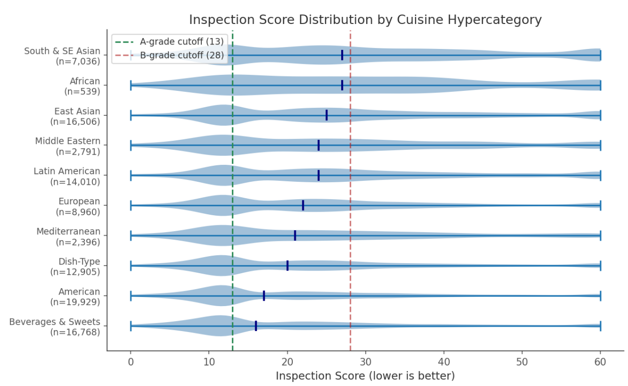


Figure 9: Score distribution by cuisine hypercategory (violin plot). Sample sizes are shown for each group. African and Middle Eastern restaurants tend to score worst; Beverages & Sweets and American establishments tend to score best. The blue median lines illustrate the cuisine effect documented in the regression.

York, New York, USA. *Emerging Infectious Diseases*, 24(12), 2164–2168. https://doi.org/10.3201/eid2412.180544

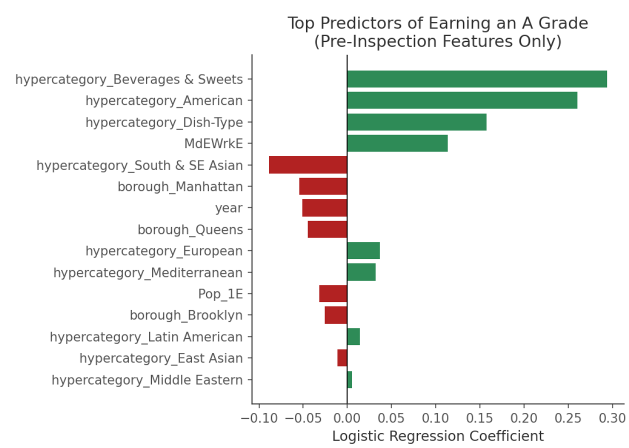


Figure 10: Top predictors from the pre-inspection logistic regression (borough, cuisine, neighborhood variables). No predictor has a large coefficient, consistent with the low AUC of 0.60 for this model.